Examining learnability of OT grammar

A view from Kashmiri loanwords

Sneha Ray Barman

Center for Linguistic Science and Technology, IIT Guwahati

Table of contents

1. Introduction
2. A Comparative overview of Kashmiri Phonology
3. OT Analysis
4. Learnability with Constraints
5. Discussion and Future Steps

Intro

Background

- **Blair and Ingram** (2003) mentioned that loanword formation provides a good environment for studying phonological structures and mechanisms because of its restrictiveness.
- **Yip**(1993), **Silverman**(1991) studied perceptual influences in Cantonese loanword phonology.
- **Choi**(2002) detailed English loanwords in Korean, **Nagarajan**(2014) studied the constraints for Bangla loanword phonology over the ages.
- Epenthesis has been deemed as the most common repair strategy for borrowed words(**Davidson et al** 2004). **Kar**(2013) showed how complex onsets are treated in regional varieties of Bangla, **Mahmood et al**(2011) examined similar strategy in the Indian variety of Punjabi.
- Aspiration of voiceless stops is also an interesting repair strategy in languages like Korean(**Hirano** 1994; **Lee** 2001).

Why Kashmiri loanwords?

- Koul(2005), Kachru(1969;2006) have intensely studied Kashmiri linguistics but there is no account of OT constraints.
- Kashmiri loanword phonology is visibly significant due to phonological as well as morpho-phonological alternations.
- It is the only Indo-Aryan language devoid of four-way contrast in plosives. Despite having unaspirated voiceless stops in native grammar, the aspiration of voiceless stops in borrowed words is an interesting case. Proposing an OT grammar is the first step to examining this phenomenon.
- Epenthesis as a repair strategy has been studied before but epenthesis in [/s/+ C] a regular case in Indian languages.

Purpose of the Study

The aim of this presentation is-

- to propose a set of OT constraints to account for the phonotactics (*specifically aspiration and epenthesis*) of Kashmiri loanword phonology, and
- to check the learnability of the proposed grammar following Tesar and Smolensky (1995)'s learning algorithm.

A Comparative overview of Kashmiri Phonology

Overview:Kashmiri native phonology

- The basic syllable structure of Kashmiri is (C)V(C). Onset and coda are not mandatory. However, the maximum syllable structure is (C) (C) V (C) (C).

Kashmiri	Gloss	Syl.struc.
/aː/	*yes*	V
/kaːh/	*eleven*	CVC
/aːmb/	*mango*	VCC
/drog/	*expensive*	CCVC
/aksar/	*often*	VCVC

Table 1: General syllable structure

Overview of Kashmiri loanword phonology

- **Aspiration of voiceless stops at coda position.**

Persian	Kashmiri	Gloss	Hindi-Urdu	Kashmiri	Gloss
/paːk/	/paːkh/	*pure*	/mulk/	/mulukh/	*country*
/caːlaːk/	/caːlaːkh/	*clever*	/paːp/	/paːph/	*sin*
/naːzuk/	/noːzukh/	*delicate*	/raːt/	/raːth/	*night*
/poːšaːk/	/poːšaːkh/	*dress*	/davaːt/	/davaːth/	*inkpot*

Table 2: Partial list of loanwords in Kashmiri (Koul 2005)

English	Kashmiri
doctor	/daːkht̥ar/
minute	/minat̥h/
rate	/reːt̥h/
paper	/peːpar/

Table 3: Partial list of loanwords in Kashmiri (Koul 2005)(contd.)

- **Insertion of vowel /ɨ/ in cosonant clusters.**

English	Kashmiri	Gloss
/eɜplem/	/pleːn/	*aeroplane*
/klɜːk/	/klərɨk/	*clerk*
/skuːtə/	/sɨkuːt̥ar/	*scooter*
/steɪ.ʃən/	/sɨt̥esaːn/	*station*
/spiːd/	/sɨpiːd/	*speed*
/skuːl/	/sɨkuːl/	*school*
/nɜːs/	/narəs/	*nurse*
/lɪp.stɪk/	/lip.sɨ.tik/	*lipstick*

Table 4: Partial list of loanwords in Kashmiri (Koul 2005)

OT Analysis

General Constraints

- **Constraints:** Well-formedness constraints:
  ONSET-A syllable must have onset.
  NO CODA- Syllables are open. (Ito 1989; Prince & Smolensky 1993)
  *COMPLEX-Cmplex onset and coda are not allowed.
- Koul(2005) explained that the word-initial consonant clusters in Kashmiri are grammatical only if a stop/affricate/fricative is followed by /r/ while word-final clusters are acceptable only if a nasal/fricative is followed by a stop.
- However, the existence of loanwords like /klaːb/ *'club'* or /pleːn/ *'aeroplane'*(Kachru 2006) suggests that the clusters follow **Sonority Sequencing Principle**(initial clusters should rise in sonority while the final clusters should fall). We use SONORITY as a markedness constraint to account for this.

- Therefore, constraint ranking: SONORITY » FAITHFULNESS » *COMPLEX,NO-CODA,ONSET

Input:/*drog*/	SONORITY	IDENT-IO	*COMPLEX	NO-CODA	ONSET
⇨ a.**drog**			*	*	*
b. dro		!*	*		*
c. dɨrog		!*		*	*

Table 5: General syllable structure in Kashmiri

Case1: Aspiration of Stops

- Analysing the data from Table 2 and 3, it is clear that voiceless unaspirated plosives (/k/,/p/,/t/) are aspirated (/kh/,/ph/,/th/) in syllable-final (/pa:kh/,pa:ph/,/minaṭ̥h/ etc.) position. The constraint to account for this instance- ***T[-vce,-sg]syll**(voicelss unaspirated plosives are not allowed syllable-finally).
- ***[+spread glottis]** is a context-free markedness constraint that disallows aspiration, and **IDENT-IO** is a faithfulness constraint that correspondents in input and output have identical features.
- Since aspiration of voiceless stops is a marked feature of Kashmiri, it is ranked as the highest constraint that violates the faithfulness constraint IDENT-IO as well as *[+spread glottis] yet outputs a grammatical element.

Constraint ranking: *T[-VCS,-SG]SYLL » IDENT-IO » *[+SPREAD GLOTTIS]

Input:/*ca:la:k*/	*T[-VCS,-SG]SYLL	IDENT-IO	*[+SPREAD GLOTTIS]
a. ca:.la:k	*!		
⇨ b. **ca:.la:kh**		*	*

Table 6: Aspiration in Kashmiri loanwords

Case2: Epenthesis

- /ɨ/ is inserted when consonant clusters violate Sonority Sequencing Principle.
- Kar (2013) stressed examining the epenthesis in clusters consisting of coronal [/s/+stop](eg. /sku:l/->/iskul/).
- Repair strategy similar to Kashmiri is seen in Lenakel Kager (1999) (/t-n-ak-ol/->/tɨ.na.gol/)

- Constraints:
- The general well-formedness constraints I propose are-
  - SONORITY- Initial cluster should rise in sonority while the final cluster falls.
  - DEP-IO- No epenthesis allowed.
  - CONTIGUITY- No medial epenthesis or deletion allowed (Kager1999; Gouskova 2002).
- Why not (ɨskul or skulɨ)?- Alignment constraints are introduced to avoid initial or final epenthesis.
  - ALIGN-L restricts epenthesis on the left edge of the Prosodic word.
  - ALIGN-R restricts epenthesis on the right edge.
- Why /sɨkuːl/ and not /səkuːl/?- /ɨ/ appears to be the epenthetic vowel with features [+HIGH,-BACK,-LOW,-ROUND]. We can count these as the four features as marked in the language.
- DEP(MF)- A surface instance of [+LOW], [+HIGH], [+ROUND], or [+BACK] must have an identical underlying correspondent.(Shademan 2002)

Proposed constraint ranking: SONORITY,ALIGN-L,DEP(MF)» ALIGN-R» DEP-IO, CONTIGUITY

Input:/spiːd/	SONORITY	ALIGN-L	DEP(MF)	ALIGN-R	DEP-IO	CONT.
a. spiːd	*!					
b.is.piːd		*!	*		*	
⇨ c.sɨ.piːd				*	*	*
d. sə.piːd			!*	*	*	*
e. ɨs.piːd	*!	*			*	*

Table 7: Epenthesis in Kashmiri loanwords

Learnability with Constraints

Basic Ideas

- **Question:** Given a set of surface forms of the target language (Kashmiri), is it possible for the learner to discover the correct constraint ranking?
- **Assumptions:** A child only has access to the optimal candidates (positive evidence) of the grammar in their linguistic environment.
- Constraint reranking strictly adheres to the violation of constraints instead of satisfaction.
- A top-ranked constraint can be violated by an optimal candidate as long it is violated at least as many times by other outputs.
- The algorithm demotes constraints in winner-marks immediately below the constraints in loser-marks. The demotion must be minimal, hence reverse ranking is suggested (Kager, 1999).

Initialization

- **Linear stratum (for candidates in table 5): H_0 : {ALIGN-L, ALIGN-R, CONTIGUITY, DEP-IO, DEP-MF, SONORITY}** .
- In the following steps, each suboptimal candidate is compared (on basis of harmony $<$) against the optimal candidate to deduce the loser and winner marks. The mark-data pairs are listed below-

Mark-data pair table

ark-data pairs (subopt $<$ opt)	Loser-marks	Winner-marks
$H_1 : e < c$	{*SONORITY,*ALIGN-L}	{*DEP-IO, *CONTIGUITY}
$H_2 : d < c$	{*DEP(MF)}	{*ALIGN-L,*DEP-IO,*CONTIGUIT
$H_3 : b < c$	{*ALIGN-L,*DEP(MF)}	{*DEP-IO}
$H_4 : a < c$	{*SONORITY}	...

Table 8: Mark-data pairs of Table 5 candidates

Performing Recursive Constraint Demotion

- H0 is the initial strata where the constraints remain unranked.
- At H1, suboptimal candidate *e*[ɨs.piːd] is less harmonious than the optimal *c*. *DEP-IO and *CONTIGUITY are demoted immediately below H0. Thus, the winner-marks constraints are demoted to a stratum immediately below the loser-marks constraints.

  {ALIGN-L, ALIGN-R, CONTIGUITY, DEP-IO, DEP(MF), SONORITY}

  »

  {DEP-IO,CONTIGUITY}
- At H2, *d* [səpiːd] is compared against the optimal candidate *c* [sɨpiːd]. The table shows that Kashmiri would rather allow right-edge insertion than ungrammatical vowel insertion.

- {ALIGN-L, DEP(MF), SONORITY}

  »

  {ALIGN-R, DEP-IO, CONTIGUITY}
- ALIGN-R already dominates DEP-IO and CONTIGUITY in H1 demotion, hence-

  {ALIGN-L, DEP(MF), SONORITY}

  »

  {ALIGN-R}

  »

  {DEP-IO, CONTIGUITY}
- Minimal demotion occurs in H3 and H4 as the undominated constraints are already highest-ranked.

Discussion and Future Steps

- Despite the fact that voiceless unaspirated stops are aspirated in the coda position, words like [šuɨp], [aksar], [kašt] exist in native grammar.
  - **Is there an underlying representation?**
- Gouskova (2002) proposed that if a /s/ is followed by a consonant, it cannot be broken up, but it is very regular in Kashmiri. The constraints are strictly motivated by S.S.P.
- However, there are examples like *clerk* >klərɨk or *park* > parɨkh where the cluster does follow S.S.P. yet is broken up.
  - **Are there other motivations for epenthesis?**
- Though the learner appears to converge into target grammar, the assumption that a real-life learner has access to underlying representation and acoustic cues is arbitrary.
- Tesar & Smolensky's (1993) algorithm is proved to converge even if any one of the constraints leading to the actual ranking of the dataset, the next intention of this research is to analyze the hidden structures if any.
  - **What happens when the learner is presented with incorrect input**
  - **Can it achieve the correct ranking given the hidden structure?**
  - **Can Maximum Entropy grammar or Stochastic OT account for these instances?**.

Questions?

Please reach out to me for feedback and suggestions at raybarmansneha2@gmail.com

References

Kager, R. (1999). *Optimality Theory*. Cambridge Textbooks in Linguistics. Cambridge University Press.

Kar, S. (2013). Complex onsets in regional varieties of bangla. *Language Sciences*, 40:212–220.

Tesar, B. and Smolensky, P. (1995). The learnability of optimality theory: An algorithm and some basic complexity results.

Acknowledgment

Thanks to the anonymous reviewers of ConSOLE 31, my supervisor Prof. Shakuntala Mahanta, and of course the viewer!

THANK YOU!